

Prafull Kumar Prajapati

Email: prajapatiprafull12@gmail.com | **Address:** Richardson, TX | **Contact No:** 945 268 5954
LinkedIn: linkedin.com/in/prafull-prajapati | **Github:** github.com/Prafull1Kumar | **Website:** prafull.tech

EDUCATION

The University of Texas at Dallas (UTD) | M.S. in Computer Science | Richardson, TX May 2026
Coursework: Database Design, Operating System, Object-Oriented Programming, Machine Learning, Software Development
Scholarship: Dean's Scholarship for Outstanding Academic Achievement

Indian Institute of Technology Kharagpur (IIT KGP) | B.Tech in Aerospace Engineering | Kharagpur, India June 2021
Coursework: Data structures and algorithms, Probability and Statistics, Computer Organization and Architecture

TECHNICAL SKILLS

- **Programming Languages:** C/C++, JavaScript, Core Java, Python, TypeScript, SQL, C#, .NET, Rust, Go, Ruby
- **Software Tools/ OS:** PowerBI, Azure DevOps, PgAdmin, Google Analytics, Redis, Docker, Linux, Git, Snowflake, PyTorch
- **Libraries/Frameworks:** Node.js, ExpressJS, FastAPI, Spring Boot, Kafka, Google Cloud, Angular, Azure, Rails, FAISS, Next.js
- **Databases & Data:** PostgreSQL, MySQL, Snowflake, MongoDB, Elasticsearch, Redis
- **LLMs/ VLMs:** LangChain, LangGraph, multi-agent workflows, Agentic AI, GenAI, Crew AI, CLIP, LLaVA, BLIP-2

PROFESSIONAL EXPERIENCE

Graduate Assistant, The University of Texas at Dallas, Richardson, United States August 2025 – Current

- Developed AI-powered applications leveraging Python, PyTorch, and Transformer-based models for natural language processing tasks.
- Implemented RAG pipelines integrating vector databases and Sentence-BERT embeddings for semantic search and question answering.
- Conducted experimentation with Vision-Language Models (CLIP / BLIP-2) for multimodal learning and research-oriented projects.

Software Dev Engineer Intern, Amazon Web Services, Seattle, United States May 2025 – August 2025

- Integrated long polling in Java based microservices to get the latest log from DynamoDB database to reduce redundant requests.
- Optimized polling by adding a configurable timeout and reducing concurrent blocked threads by asynchronous request-reply pattern
- Achieved 84% reduction in total requests, which significantly results in lowering CPU, memory, file descriptor usage, and network I/O.

Backend Team Lead, LivSYT, Hyderabad, India June 2021 – July 2024

- Created a strategy to migrate AWS to Azure Services and also managed CI/CD pipelines for numerous services using Azure DevOps.
- Designed architectures while considering Azure costs, recommend suitable database infrastructure, and achieved a 25% cost reduction.
- Identified the REST APIs that require microservices and implement them in TypeScript to reduce the Lambda cold start time by 30%.
- Created a robust mechanism to locate rejected mails and resend them by Simple Queue Service (SQS), reducing rejected mails by 60%.

Software Developer, Arth, Hyderabad, India January 2018 – May 2021

- Built an API Gateway to securely pass user parameters and trigger the AWS Lambda for sending emails by Simple Email Service (SES)
- Implemented scalable unit test infrastructure using in-memory database integration in Loopback framework for REST API testing
- Designed and implemented .NET applications on Azure cloud, leveraging microservices and containerization to achieve 99.9% uptime
- Efficiently used Sequelize with NodeJS and deployed microservices using the Serverless Application Model on AWS Lambda.

PROJECTS/HACKATHONS

AI Job Outreach Agent with LLM + RAG Platform, The University of Texas at Dallas February 2026 – Current

- Built an agent-based LLM system that analyzes job descriptions and recruiter profiles to automate personalized outreach emails.
- Implemented RAG pipeline with Sentence-BERT embeddings and FAISS to retrieve resume content and ensure grounded responses.
- Designed multi-step agent workflow for job parsing, resume evidence retrieval, email drafting, and recruiter outreach automation.
- Integrated FastAPI backend and Gmail API to generate and send recruiter-targeted emails with structured prompts and context retrieval.

Custom Distributed Search Engine, The University of Texas at Dallas August 2025 – December 2025

- Built a distributed search engine using Apache Spark (PySpark) implementing TF-IDF indexing for scalable information retrieval.
- Designed data pipelines using Hadoop MapReduce for large-scale data refinement and web graph construction across 100K+ webpages.
- Designed React interface to display custom engine results side-by-side with Google and Bing, enabling clear comparison of searches.
- Applied clustering and ranking techniques to group related pages and improve search relevance and query performance.

OnboardIQ (Goldman Sachs Problem Statement), HackUTD 2025 November 2025 – November 2025

- Built an onboarding banking platform with an ML/AI-driven loan approval workflow that keeps the process fast, with dual-theme UI.
- Automated KYC by streaming ID uploads in S3 by invoking AWS Textract OCR, and surfacing status directly in vendor review page.
- Adapted a Kaggle Machine Learning model with auto-reject thresholds to pre-score loan and store the confidence score of the workflow
- Built a full-stack web app using Node.js and PostgreSQL (Amazon RDS) on the backend, deployed on EC2 with a React frontend.

ACTIVITES

Leetcode coding contest: Ranked in the top 1% of candidates in bimonthly LeetCode competitive programming challenge August 2023

Publicis Sapient JumpStart: Ranked among the top 0.5% in Publicis Sapient JumpStart 2022 with a score of 347 October 2022